Does Higher Government Debt Go with Slower Growth the Next Year? Evidence From 191 Economies, 2014–2023

Gachy Ultimo · ECON 3916 Midterm Project · November 9, 2026

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Question	Across 191 economies from 2014 to 2023, is higher general government gross debt (% of GDP) in the previous year associated with slower growth in real income per person, holding last year's income and each country and each year fixed?
Hypothesis	Heavy debt should slow growth due to higher interest costs and less room for private borrowing and investment, so I expected a negative coefficient on previous-year debt.
Headline finding	A country-year that follows a year with debt 10 points of GDP higher has income growth 0.20 percentage points higher (95% CI −0.11 to +0.51 points, N = 1,901 country-years, 191 economies). The interval includes zero, so the hypothesis is neither supported nor rejected. It is a conditional within-country association, not a causal effect. With same-year debt the sign reverses, which mostly reflects growth moving the debt ratio.
Data	IMF World Economic Outlook (WEO), version 9.0.0 (October 2026), downloaded 2026-11-02 from IMF Data Explorer. One row is one economy in one year. Main sample: 1,901 country-years from 191 economies, 2014–2023.
Method	OLS of yearly income-per-person growth on previous-year debt and the log of previous-year income, with country and year dummies. Standard errors are clustered by country, with t-based intervals.

1. The Question and the Data

Whether public debt holds back growth matters to finance ministries, to the IMF and the rating agencies that advise on fiscal limits, and to citizens in any budget debate. Comparing countries answers it badly, because high-debt countries differ from low-debt ones in many ways that also affect growth. I ask a narrower question the data can answer: when a country enters a year with more debt than is usual for that country, does it grow less that year than other countries do in the same year?

The outcome is the yearly percent change in real GDP per person, in constant purchasing-power-parity dollars (IMF series NGDPRPPPPC). I calculated it as this year's level divided by last year's, minus one, times 100. The main explanatory variable is general government gross debt as a percent of GDP (GGXWDG_NGDP), measured in the previous year. The confounder is the log of previous-year income per person: poorer countries tend to grow faster (they catch up) and to carry different amounts of debt. Investment (gross capital formation, % of GDP, NID_NGDP) is the third variable, which I run as a check and not in the main model (Section 3).

The data are a product of measurement decisions. IMF staff compile the World Economic Outlook twice a year from figures reported by national authorities and from their own estimates, to monitor and forecast economies, not to test this hypothesis. Three decisions matter here. First, "general government" debt covers different levels of government in different countries (state and local governments and social security funds are included in some), so the numbers are not perfectly

comparable. Second, income is in constant 2021 international dollars, which makes levels comparable across countries and years but depends on the price survey behind it. Third, some recent values are IMF estimates and not statistics: of 5,578 debt, income, and investment values for 2014–2023, 20 (0.4%) fall after the IMF's own "latest actual data" year for that series, and 66 of the 563 series have no such year recorded, so I cannot classify their values.

Preparing the file changed it in four ways. The download is the whole WEO database (8,202 rows and 79 columns: one row per economy and indicator, one column per year, 1980 to 2031). I dropped the IMF's regional and group rows (World, euro area and so on, with codes such as G110) so that every row is one economy; the values arrived as numbers, with empty cells for missing values. I kept the three indicators and reshaped the year columns into one row per country-year (2,167 rows, 197 economies, 2013 to 2023). Finally, I copied each economy's previous-year income and debt onto the current year's row. The 2013 rows only supply "last year" for 2014; the models use 2014–2023.

Missing data are not random. The main model needs growth, previous-year debt, and previous-year income, and 1,901 of the 1,970 country-years in 2014–2023 have all three. Six economies drop out entirely: Macao, Somalia, Syria, Venezuela and Yemen report no debt in any year, and Eritrea and Syria have no income growth (Syria has neither). Investment is missing in every year for 24 economies, mostly microstates (Aruba, Andorra, Kiribati, Liechtenstein, Tuvalu) and fragile states (Iraq, Lebanon, Liberia, Sudan, Syria, Yemen), which is why I keep it out of the main sample. These economies are poorer: the median income of the economies with no debt data is about $5,500, against about $16,100 for the rest, and about $6,600 against $16,300 for those with no investment data.

I label the gaps as not MCAR, because they cluster in small and poor economies and in conflict cases, so missingness depends on observable traits. For the microstates, MAR given income and size is the label I can defend, since coverage plausibly depends on a country's size and statistical capacity, both of which I observe. For the conflict cases (Syria, Yemen, Somalia, Venezuela) I cannot rule out MNAR: a government may stop reporting when conditions collapse, and collapse is tied to the debt and growth values I never see. I cannot test that. The consequence is the same either way: my sample is the economies with regular national accounts, and my estimate says nothing about the most fragile ones. Table 1 summarizes the main sample.

Table 1. Summary statistics, main sample (growth and investment in the same year, debt and income from the previous year)

Variable (units)	Mean	Median	SD	Min	Max	N
Income per person growth (% a year)	1.56	1.90	5.22	−34.02	59.29	1,901
General government gross debt, previous year (% of GDP)	55.04	47.49	35.73	0.05	357.45	1,901
Investment (% of GDP)	24.40	23.38	8.35	−15.68	75.92	1,691
Income per person, previous year (constant 2021 intl. $)	24,753	15,414	26,118	830	175,785	1,901

Source: IMF World Economic Outlook, version 9.0.0, downloaded 2026-11-02. Income is real GDP per person in constant 2021 international dollars. Investment is reported for 1,691 of the 1,901 country-years, so its N is smaller.

2. Descriptive Evidence

Figure 1 shows the outcome. Growth is centered on a median of 1.90% a year (mean 1.56%) and is close to symmetric, with a skewness of 0.20, so I model it in levels. A log would also fail, because growth is negative in some years. The largest value is 59.3% (Guyana in 2022), and the smallest is −34.0% (the Maldives in 2020), which is why Section 4 checks influential observations.

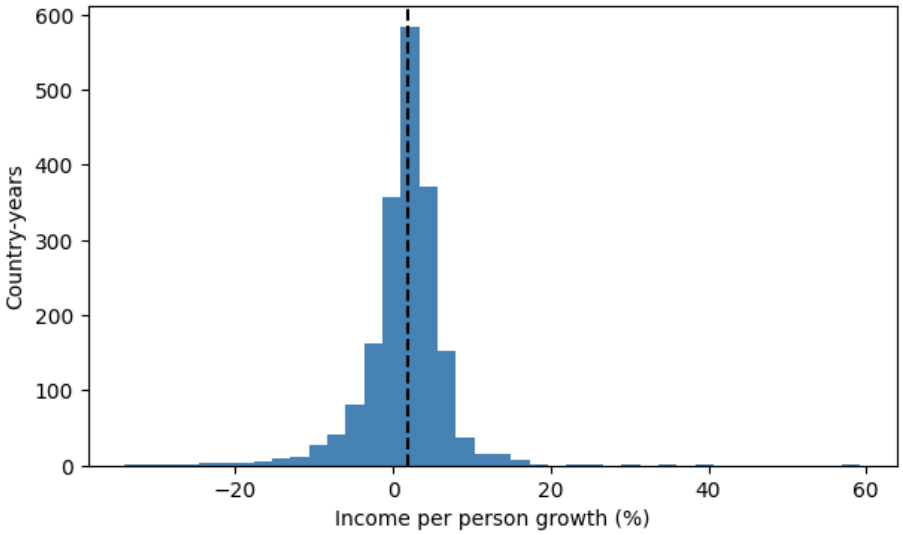


Figure 1. Growth in income per person is centered near 1.9% a year and is roughly symmetric (skewness 0.20), with thin tails from crisis and boom years. Source: IMF WEO 9.0.0; author's calculation.

Figure 2 plots growth against previous-year debt before any controls. There is almost no pattern: the correlation is 0.024. Most country-years have debt below 100% of GDP, and the few above 150% sit on both sides of zero growth. This raw picture compares different countries and different years, so it can mislead in either direction; the model in Section 3 compares each country with itself and sets aside what hit every country each year, such as COVID in 2020.

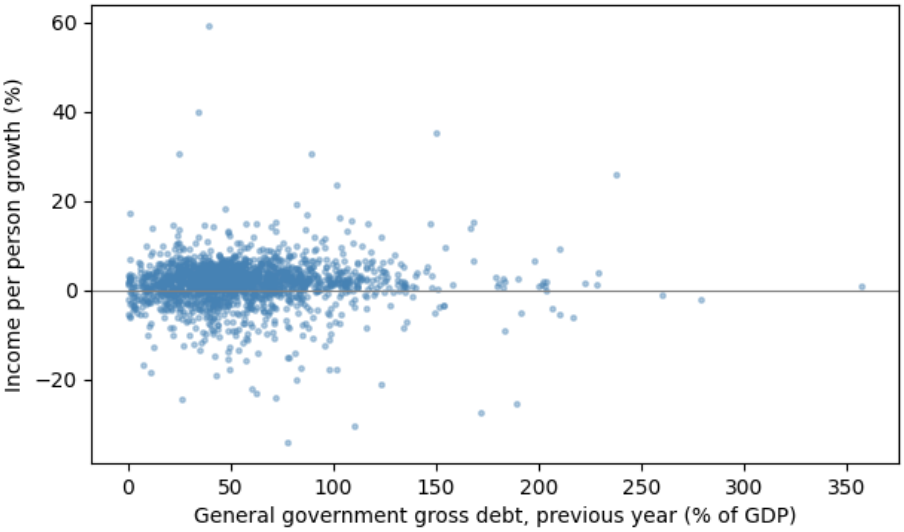


Figure 2. Before any controls, previous-year debt and growth show almost no relationship (correlation 0.024); most country-years have debt below 100% of GDP, and the few above 150% scatter on both sides of zero. Source: IMF WEO 9.0.0.

In 2020, average growth fell from 1.7% to −5.7% and average debt jumped from 54.8% to 66.5% of GDP in the same year; in 2021, growth rebounded to 4.9% while debt was still high (64.1%). Within a year, debt and growth move in opposite directions, which is what makes same-year debt misleading. Across years, high debt in one year is followed by a rebound in the next, which is one reason a lagged coefficient can turn positive. Year dummies absorb the average 2020 shock, but not the differences in how hard each country was hit.

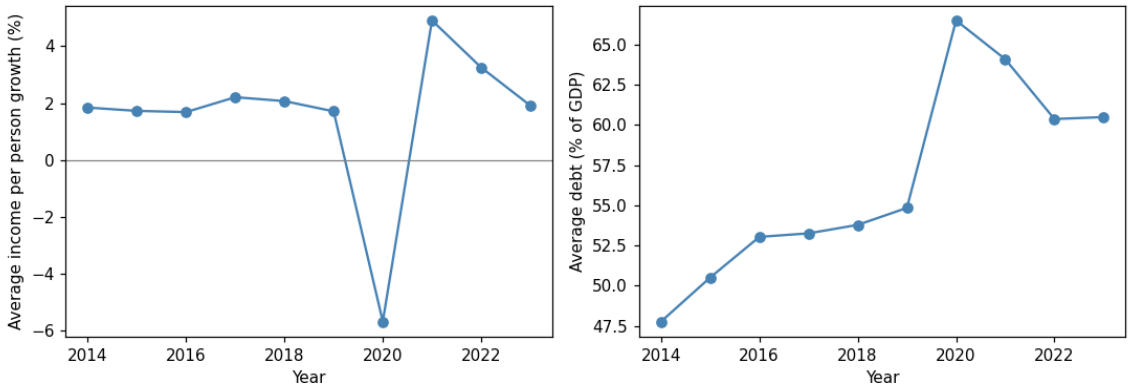


Figure 3. Average growth fell from 1.7% in 2019 to −5.7% in 2020 and rebounded to 4.9% in 2021, while average debt rose from 54.8% to 66.5% of GDP in 2020 and stayed above 60% afterwards. Source: IMF WEO 9.0.0; main sample.

3. The Model

I fit the model below by OLS. Here i is an economy and t a year; α_i are country dummies (one per economy), and γ_t are year dummies (one per year). Standard errors are clustered by country, because the ten rows of one country are not ten independent observations, and intervals use the t distribution. The full results are in Table A1; column 2 is the main model.

$$Growth_{it} = \beta_0 + \beta_1 \, Debt_{i,t-1} + \beta_2 \, ln(Income_{i,t-1}) + \alpha_i + \gamma_t + \varepsilon_{it}$$

I use previous-year debt as the main variable because same-year debt as a share of GDP moves mechanically with growth: a bad year shrinks GDP, which raises the ratio in that same year. Lagging removes that channel, but it still doesn't make the estimate causal.

The coefficient on previous-year debt is +0.020 (standard error 0.016): one more percentage point of debt/GDP in the previous year goes with 0.02 percentage points more income growth this year, or 0.20 points for a 10-point difference. The 95% interval runs from −0.011 to +0.051 per point (−0.11 to +0.51 points for 10 points). Since the interval includes zero, the data cannot distinguish this from no relationship, but they do not rule out a small positive or a small negative one. Because of the country dummies, this compares each economy with itself: it is the difference in growth in years that follow a year when the country's debt was above its own average, relative to other countries in the same year. That is a different question from whether high-debt countries grow more slowly.

The coefficient on the log of previous-year income is −9.12 (standard error 4.92). An income level 1% higher last year is associated with growth 0.09 points lower this year, which fits poorer economies catching up. The 95% interval runs from −18.8 to +0.6 (−0.19 to +0.01 points per 1%), so the evidence for catch-up within countries is borderline.

Poorer countries tend to grow faster and to carry different amounts of debt. It is measured in the same year as the debt I use, so earlier debt may have helped shape it; if so, controlling for it removes part of the effect of debt, which is the main way it could be a bad control. I keep it because catch-up is the obvious rival explanation, and column 1 shows the debt coefficient without it (+0.039, against +0.020 with it). The country dummies block confounding from anything fixed about a country (institutions, geography, culture), and the year dummies block shocks that hit every country at once, such as 2020. Clustering leaves the coefficient alone and widens the standard error to reflect repeated rows from the same country. If heavy debt crowds out investment and that slows growth, controlling for investment removes part of the effect I am asking about. That is why it is not in the main model. Columns 3 and 4 compare the model without and with investment on the same 1,691 country-years; the debt coefficient moves from +0.037 to +0.041, so adding investment barely matters.

Restricting the main model to the 1,691 country-years that report investment raises the debt coefficient from +0.020 to +0.037, because the economies without investment data are small and fragile. And using same-year debt (column 5) reverses the sign: −0.063 (standard error 0.015), so 10 points more debt goes with 0.63 points less growth in the same year. I took lagged debt as the main model on my instructor's advice and for the reason above, and I left investment out because of the mediator argument. Table A1 reports all of them. The R^2 of 0.46 is mostly the dummies: country and year dummies alone give 0.44.

4. Diagnostics

Figure A1 in the appendix plots the residuals against the fitted values (left) and against previous-year debt (right). The residuals are centered on zero, with a handful of large positive residuals (up to about 45 points). The spread is wider where fitted growth is far from the middle, and the extremes in the right panel point to a few extreme observations, which the next checks quantify.

The Breusch–Pagan test on all regressors, dummies included, rejects constant variance (LM = 721.1, $p \approx 1\times10^{-59}$). With only debt and income as regressors, it is borderline (LM = 5.01, p = 0.081). All these values indicate that the spread of the errors differs across countries and years, with even larger swings in fragile economies and during the COVID years. So, I used robust standard errors. The table already clusters by country, which is robust to unequal variance, and the choice does not change the conclusion. The standard error on debt is 0.0157 clustered and 0.0151 with HC3, and both intervals include zero.

The variance inflation factors, with the dummies included, are 9.28 for previous-year debt and 274.5 for log previous-year income. An income level is mostly a country's own level plus a trend shared by all countries, so little independent movement is left. That is why its standard error is wide (4.92). Debt and income are not tangled with each other (correlation 0.144), and the debt figure is below 10, so collinearity does not explain the inaccurate debt estimate.

Using Cook's distance with the cutoff 4/n = 0.0021, 94 of the 1,901 rows stand out. A single-row country (Seychelles) has an undefined distance because its own dummy fits it exactly, so I leave it out of the ranking. The largest values are Guyana 2022 (growth 59.3%), Guyana 2020, the Maldives 2020 (−34.0%), Libya 2017, Guyana 2023, the Maldives 2021, Libya 2021, and Sudan 2023. Without the five most influential rows, the coefficient is +0.018 (standard error 0.016, N = 1,896). Without all 94 rows above the cutoff, it falls to +0.002 (standard error 0.006, N = 1,806). Either way, it stays small and indistinguishable from zero, but the drop shows that part of the small positive estimate comes from extreme boom and bust years.

COVID: Because 2020 hit growth and raised debt in almost every country at once, I also re-estimated the main model without 2020 and 2021 (coefficient +0.010, standard error 0.016, N = 1,521) and on 2014–2019 only (+0.001, standard error 0.023, N = 1,140). Both are smaller than the full-sample estimate, and both intervals include zero, so the positive sign in the main model depends on the pandemic years.

5. Threats and Conclusion

For the coefficient to be causal, nothing else that moves growth could change within a country along with its previous-year debt, once last year's income, the country, and the year are held fixed. Fiscal policy, financial crises, and interest rates can move debt and growth together, and last year's income, my control, also could be partly shaped by earlier debt. Another threat is that a slump last year raises last year's debt ratio and predicts a rebound this year, so lagged debt can pick up the rebound; the COVID check in Section 4 has some indications of this. General government covers different levels of government in different countries, and some IMF values are estimates. The sample also leaves out the most fragile economies, so the result describes economies with regular statistics.

My estimate is a conditional within-country association, not the effect of debt on growth: the difference in this year's income growth between a country-year that follows a year of higher debt and one that follows a year of lower debt, in the same country, relative to other countries in the same year. With same-year debt, the sign reverses (−0.063), which shows how easily a mechanical link between the two can be mistaken for an effect.

I found no reliable association between last year's debt and this year's growth. The main coefficient is +0.020 points of growth per point of debt/GDP (95% CI −0.011 to +0.051; N = 1,901 country-years, 191 economies), and it shrinks toward zero without the pandemic years or the most influential rows. A reader can be confident that debt is not strongly associated with slower growth the next year, but the data do not settle the sign, because the interval includes zero. With another month, I would measure debt as central government debt (where available), extend the sample before 2014 to include more crisis and recovery years, and add interest rates and fiscal balances as controls, running the model with and without them because debt could affect both.

References

International Monetary Fund. World Economic Outlook database, version 9.0.0 (published 13 October 2026). IMF Data Explorer. https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.RES:WEO(9.0.0). Retrieved 2 November 2026. Indicators used:

- Gross debt, General government, Percent of GDP (GGXWDG_NGDP)

- Gross domestic product (GDP), Constant prices, Per capita, purchasing power parity (PPP) international dollar, ICP benchmark 2021 (NGDPRPPPPC)

- Gross capital formation, Percent of GDP (NID_NGDP)

Appendix

Table A1. Income-per-person growth (%) regressed on previous-year debt and controls

	(1) Debt only	(2) Main	(3) Main, rows with investment	(4) Adds investment	(5) Same-year debt
Debt, previous year	0.039	0.020	0.037	0.041	
	(0.018)	(0.016)	(0.019)	(0.017)	
Debt, same year					−0.063
					(0.015)
Log income, previous year		−9.116	−8.073	−8.355	−12.923
		(4.918)	(5.498)	(5.575)	(5.216)
Investment (% of GDP)				0.065	
				(0.028)	
Country dummies	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes
N	1,901	1,901	1,691	1,691	1,901
R^2	0.45	0.46	0.48	0.48	0.48

Note: Dependent variable: yearly growth in real GDP per person (%, constant 2021 international dollars). OLS, standard errors in parentheses, clustered by country (t-based). Every column includes country and year dummies. Column 2 is the main model. Column 3 re-estimates it on the 1,691 country-year s that also report investment, so that column 4, which adds investment, compares the same rows. Column 5 uses this year's debt instead of last year's. Source: IMF WEO 9.0.0.

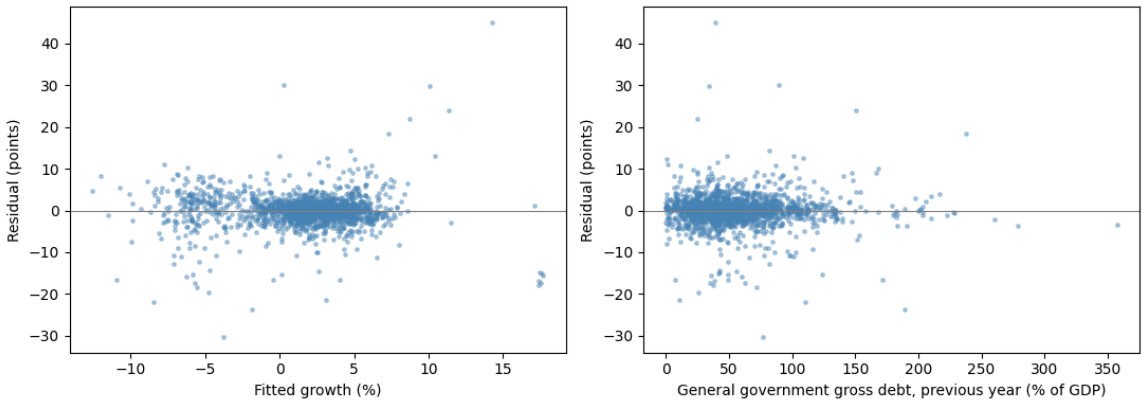


Figure A1. Residuals of the main model against fitted growth (left) and previous-year debt (right): no curve, a few large positive residuals, and a wider spread at extreme fitted values.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the data location and the order to run the cells.